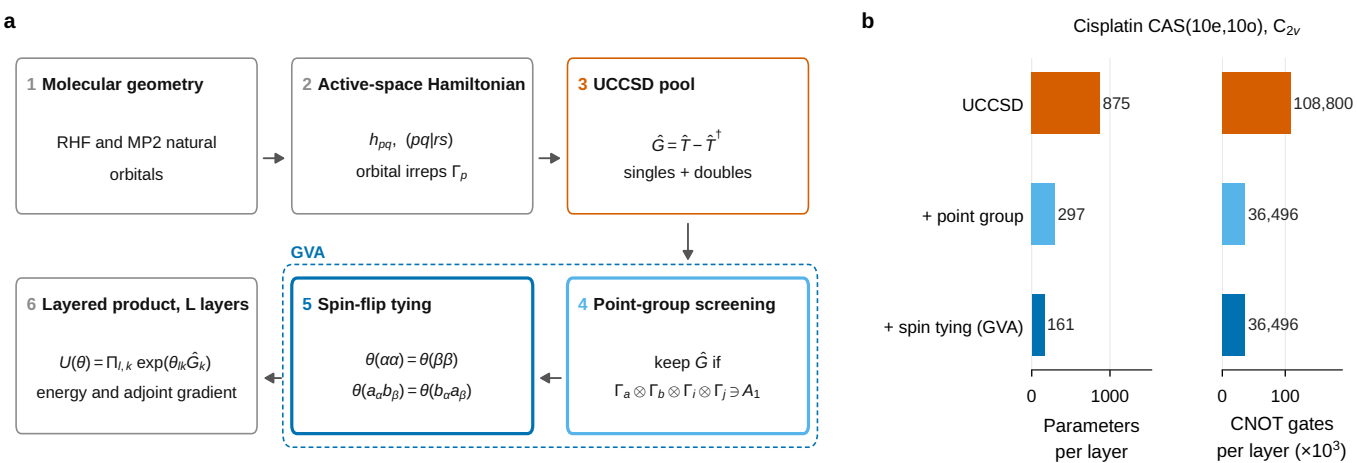
a
1 Molecular geometry
RHF and MP2 natural orbitals
2 Active-space Hamiltonian
h_{pq}, $(pq|rs)$
orbital irreps Γ_p
3 UCCSD pool
$\hat{G} = \hat{T} - \hat{T}^\dagger$
singles + doubles
GVA
6 Layered product, L layers
$U(\theta) = \Pi_{l,k} \exp(\theta_{lk} \hat{G}_k)$
energy and adjoint gradient
5 Spin-flip tying
$\theta(\alpha\alpha) = \theta(\beta\beta)$
$\theta(a_\alpha b_\beta) = \theta(b_\alpha a_\beta)$
4 Point-group screening
keep $\hat{G}$ if
$\Gamma_a \otimes \Gamma_b \otimes \Gamma_i \otimes \Gamma_j \ni A_1$
b
Cisplatin CAS(10e,10o), C_{2v}
UCCSD
+ point group
+ spin tying (GVA)
875
297
161
108,800
36,496
36,496
0
1000
0
100
Parameters per layer
CNOT gates per layer (×10^3)